Jack Rich

One Bad Apple: How Does the Success of a Fungicide Resistance Management Strategy Respond to a Proportion Of Growers Who Do Not Follow Recommendations?

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Jack Rich 02/05/2023

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1 Summary

This dissertation addresses the question of how the spatially heterogeneous usage of fungicide to treat the disease Septoria tritici blotch (STB) of winter wheat, affects the rate of fungicide resistance selection in the population of the causative fungal pathogen *Zymoseptoria tritici* (*Zt*), over multiple seasons. Many fields of winter wheat are grown in close proximity over agricultural areas with the major source of *Zt* primary inoculum initiating the STB epidemic of a season being widely dispersed airborne ascospores. Hence the rate of selection of fungicide resistance in the *Zt* population, over many years, depends on the fungicide usage of all growers over an area.

Spatially implicit and explicit models of the selection of fungicide resistance over multiple seasons incorporating between-season ascospore dispersal and heterogeneous fungicide usage are derived after dividing growers into two groups based on the dose of fungicide they apply. In the spatially implicit case, key parameters include the proportion of primary inoculum originating from outside of the field (κ) and the proportion of high concentration fungicide programmes (q). I have shown that high concentration fungicide programmes are highly influential to the rate of resistance selection in nearby fields with low concentration fungicide programmes even for low κ and $q < 0.5$. In addition, the dependence of the effective lifetime of fungicide for low concentration fungicide programmes on κ is nonlinear and exhibits a minimum for $q < 0.5$. In the spatially explicit case, a real host landscape with a parameterised inverse power law dispersal kernel is used to quantify the dependence of the proximity to and field area of high concentration fungicide programmes, on the effective fungicide lifetime in fields with low concentration fungicide programmes and this is interpreted in terms of κ.

2 Abbreviations

Table 1: List of abbreviations

Zt	Zymoseptoria tritci
STB	Septoria tritici blotch
IDM	Integrated disease management
q	Proportion of high concentration fungicide programmes
κ	Proportion of primary inoculum originating from outside of the field
λ	Scale parameter of the inverse power law dispersal kernel (in km)
α	Shape parameter of the inverse power law dispersal kernel
ρ	Proportion of resistance
s	Susceptible (subscript)
r	Resistant (subscript)

For a full list of parameters used in the within-season model see Table 2 in the Methods section.

3 Introduction

3.1 Winter wheat in the UK and *Zymoseptoria tritici*

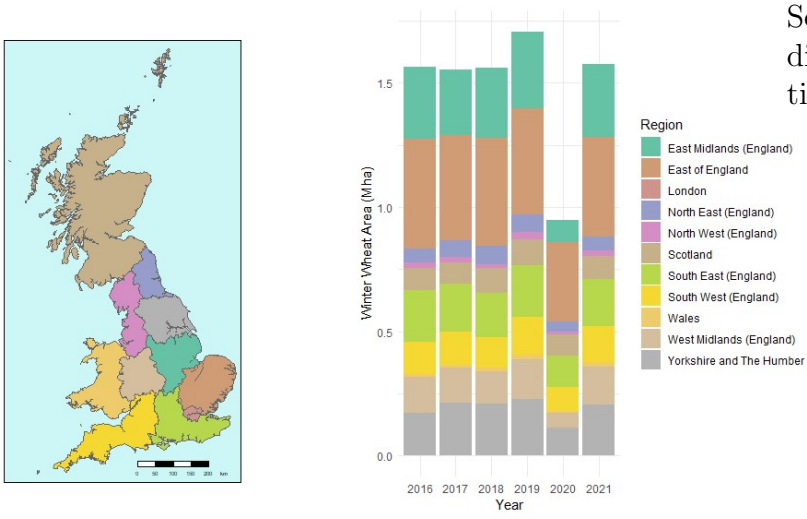


Figure 1: Bar chart showing the regional distribution of the annual area used to grow winter wheat across the UK for 6 consecutive years 2016-2021. based upon UKCEH Land Cover® Plus: Crops © [2021] UKCEH. © RSAC. © Crown Copyright 2007, Li- cence number 100017572



Figure 2: A leaf of winter wheat with Septoria tritici blotch. Black pycnidia (fruiting bodies) on brown necrotic tissue.(Ponomarenko et al., 2011)

Winter wheat are cultivars of wheat (*Triticum aestivum*), sown in the autumn to germinate and develop into young plants that remain in a vegetative state over winter resuming growth in early spring. Winter wheat is widely grown across the UK with over 1.5 million hectares of land used to grow winter wheat in a normal growing season (see Figure 1). The success of winter wheat crop is hence of vital importance to the UK food industry and economy.

The ascomycete fungus *Zymoseptoria tritici* (*Zt*) is the most damaging endemic fungal pathogen of winter wheat in the UK, causing the disease septoria tritici blotch (STB) of winter wheat. The pathogen reduces the green leaf area, through the formation of lesions (see Figure 2), thereby reducing photosynthesis causing significant annual yield losses and decrease in grain quality across the UK. Fungicides are currently a vital aspect of disease control to ensure food security, however, the effectiveness of chemical control is continually under threat by emerging fungicide resistant *Zt* strains. Hence the performance of fungicide programmes are monitored every year. Groups such as the Fungicide Resistance Action Group (FRAG-UK) provide up to date information on the resistance status, risk and suggested usage of key fungicides for the treatment of STB for growers. This advice is based on a combination of laboratory experiments, field trial data and epidemiological modelling.

Current advice is to adopt integrated disease management (IDM) to control diseases such as STB. This involves integrating control methods such as cultural, mechanical, biological and chemical control. The aim of IDM is to control disease with minimised dependence on chemical control. In

the context of STB, disease resistant and tolerant cultivars, rotations, optimised sowing timings as well as removing crop debris from previous seasons can all contribute to the integrated disease management of STB. Reducing the usage and hence dependence on fungicides to control STB, means reducing the selection for fungicide resistance, increasing the effective lifetime of key fungicides to control disease as well as reducing the impact of fungicides on local biodiversity in particular ecologically vital invertebrates such as pollinators (Schuhmann, Schmid, Manzer, Schulte, & Scheiner, 2022).

As the methods of disease control adopted by growers shifts from fungicide dependence to IDM, we may expect a higher degree of fungicide usage heterogeneity. In this dissertation growers are divided into two groups based on the dose of fungicide they apply, a either high or low concentration of fungicide.

3.2 Modelling the fungicide resistance in *Zymoseptoria tritici*

The evolution of fungicide resistance in *Zt* can be split into three phases, emergence, selection and saturation. This dissertation focuses on the selection phase. Three epidemiological principles governing the selection of fungicide resistance in populations of plant pathogens are proposed in (Milgroom, 1988). The three governing principles state that fungicide resistance will build up more slowly if:

1. The initial frequency of resistance is reduced.
2. The apparent infection rates of both fungicide-resistant and fungicide-sensitive genotypes (r_R and r_S respectively) are reduced.
3. r_R is reduced relative to r_S.

Subsequent research has focused on finding and validating principles to guide fungicide-resistance management tactics (Bosch, Oliver, Berg, & Paveley, 2014). For a single fungicide, principles have been found to help to understand the role of changing the dose (van den bosch 2011), the fungicide application timing (van den Berg, van den Bosch, & Paveley, 2013), whether treatment is applied to leaves or on the seed (Kitchen, van den Bosch, Paveley, Helps, & van den Berg, 2016), the number of sprays per seasons (Berg, Paveley, & Bosch, 2016) and more generally for multiple fungicides, whether mixtures outperform alternations (Elderfield, Lopez-Ruiz, van den Bosch, & Cunniffe, 2018)

In most of the fungicide resistance management literature for *Zt*, only within-field dynamics are considered with an exception being (Parnell, van den Bosch, & Gilligan, 2006). Splash-borne pycnidiospores (asexual) dispersed over a short range drive the epidemic within a field over a season (Karisto, Suffert, & Mikaberidze, 2022). Hence, considering only within-field dynamics is suitable over the time scale of a season. However, fungicide resistance management is ultimately optimised over a season with the expectation that the frequency of fungicide resistance will remain low for a larger number seasons. This is not necessarily the case when there is a proportion of growers that do not follow recommendations. For example it is well known that widely dispersed ascospores (sexual) are the main source of primary inoculum initiating the epidemic each year (Shaw & Royle, 1989).

In (Parnell et al., 2006) a spatially implicit model is used to identify a number of thresholds that determine the long-term regional-scale outcome of a pathogen population following the widespread and heterogeneous use of a fungicide. However, the results depend on the key assumption that there is a fitness cost to resistance. Another assumption is the homogeneous mixing of spores within the system. In this dissertation, a fitness cost of the resistant pathogen strain is not included - it is not necessarily the case that fungicide resistance in fungal pathogens has a fitness cost (Karaoglanidis, Luo, & Michailides, 2011). Additionally in the spatially explicit model there is an inhomogeneous mixing of ascospores between seasons through the use of a parameterised dispersal kernel.

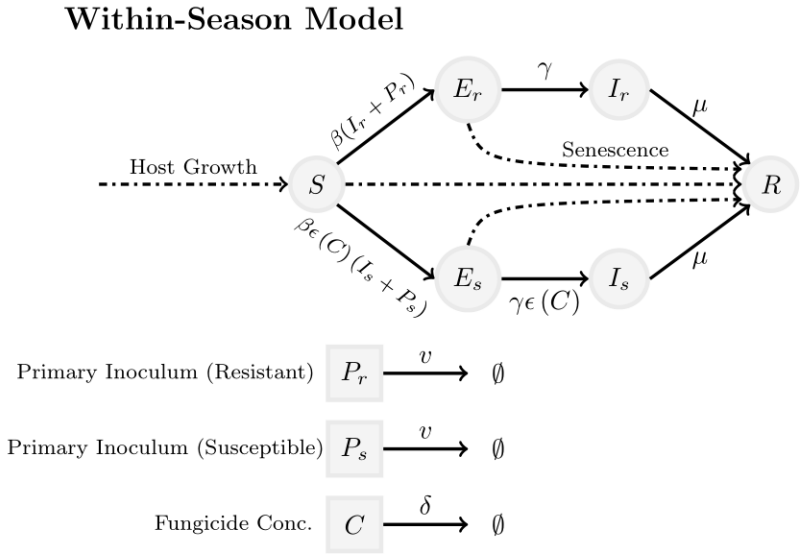
In this dissertation, a single fungicide version of the within-season model used in (Elderfield et al., 2018), based upon (Hobbelen, Paveley, Fraaije, Lucas, & van den Bosch, 2011), is extended to a spatially implicit model and a separate spatially explicit model, both incorporating between-season ascospore dispersal as the source of primary inoculum. Growers are separated into two groups based on the dosage of fungicide they apply - high or low. In the spatially explicit case a real host landscape is used with a parameterised dispersal kernel. Both models are used to develop insights into how between-season ascospore dispersal effects the rate of resistance development for high and low concentration fungicide programmes.

4 Results

4.1 Single Field Model

To model the STB epidemic within a single field of winter wheat, a compartmental susceptible-exposed-infected-removed model tracking the leaf area index (LAI), the area of leaf per area of ground is used (see Figure 3 and Methods). The model is based on the model derived, parameterised and tested against field data in (Hobbelen et al., 2011) and modified in (Elderfield et al., 2018) and simplified to account for a single fungicide.

Figure 3: Compartmental Model tracking the Leaf Area Index (LAI) of categories of leaf tissue corresponding to stages of STB. Disease can be due to fungicide susceptible and resistant *Zt*, subscript s and r respectively

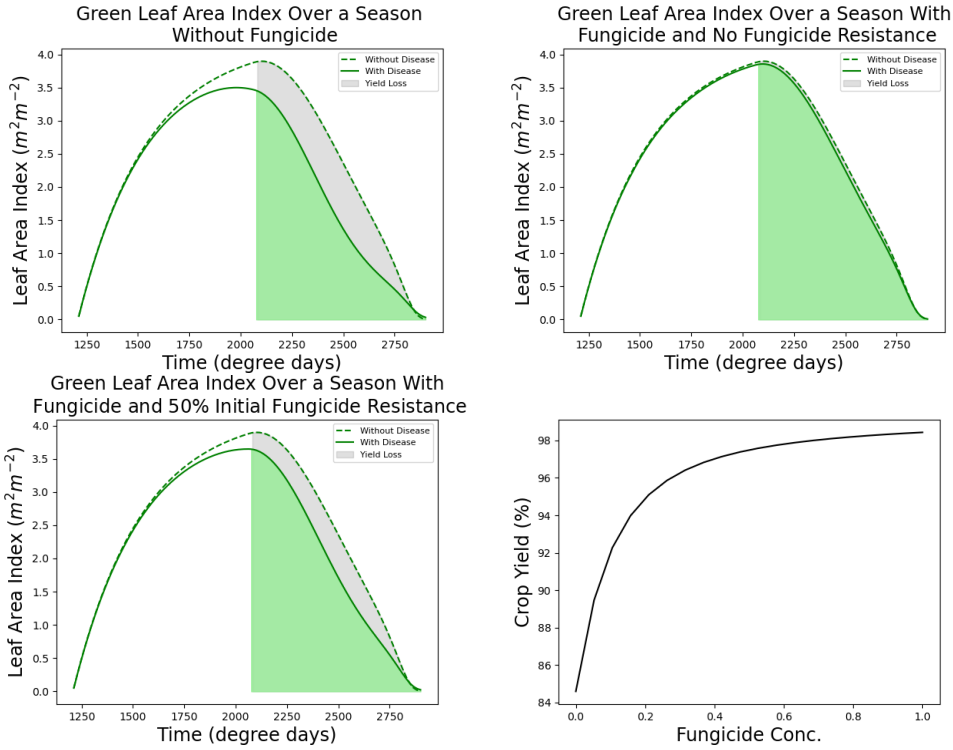


This model is used to understand the effect of fungicide concentration sprayed ($C \in [0, 1]$) on the crop yield (Y) and the proportion of fungicide resistance in the *Zt* population (ρ) over a growing season. ρ is calculated as in equation (1) and crop yield is calculated according to (2), as in (Hobbelen et al., 2011), and given as a percentage of a disease-free years yield (%).

$$\rho = \frac{I_r}{I_s + I_r} \tag{1}$$

$$Y = \frac{\int_{T_{GS61}}^{T_{GS87}} (S + E_s + E_r)\, dt}{\int_{T_{GS61}}^{T_{GS87}} S_{\text{disease-free}} dt} \times 100\% \tag{2}$$

Figure 4: Green Leaf (Susceptible) Area Index over a season 1) Without fungicide (top left) 2) With a full dose of fungicide ($C = 1$) (top right) 3) With a full dose of fungicide ($C = 1$) and 50% initial fungicide resistance 4) The relationship of crop yield and applied fungicide concentration



Over a season, the green leaf area rises until the onset of senescence and begins to fall (see Figure 4). To maximise crop yield and grain quality, green leaf area index should be maximised over the yield forming period between between growth stages GS61 and GS87 (2066 and 2900 degree days respectively). Due to STB, the senescence process is accelerated by disease induced leaf tissue necrosis resulting in lower green leaf area index over the yield forming period. Fungicide can be applied to reduce the STB epidemic size leading to a higher green leaf area index over the yield forming period. Increasing the concentration of fungicide sprayed increases crop yield and this relationship is concave and monotonically increasing (increasing the concentration of fungicide provides diminishing yield increases) (see Figure 4, bottom right). As the population of *Zt* evolves fungicide resistance, the reduction in the epidemic size possible with fungicide reduces until the entire population of *Zt* is resistant and the fungicide no longer provides any yield increase.

The model can be run over successive years, keeping track of the proportion of fungicide resistance (ρ) (see Figure 5 and Methods). The proportion of resistance (ρ) grows logistically over time shown by the linear growth of logit(ρ) over time (see Figure 5, right). The dependence of the rate of the logisitic growth of the proportion of resistance (ρ) on the concentration of fungicide applied is monotonically increasing and concave (see Figure 6). As the proportion of resistance reaches 1/2, yield sharply falls as the fungicide no longer reduces the STB epidemic size due to the dominance of the fungicide resistant strain in the population of *Zt* (see Figure 5).

Figure 5: The dependence of crop yield (%) over time (left) and the proportion of resistance (ρ) (right) on concentration of fungicide for $C \in [0, 1]$.

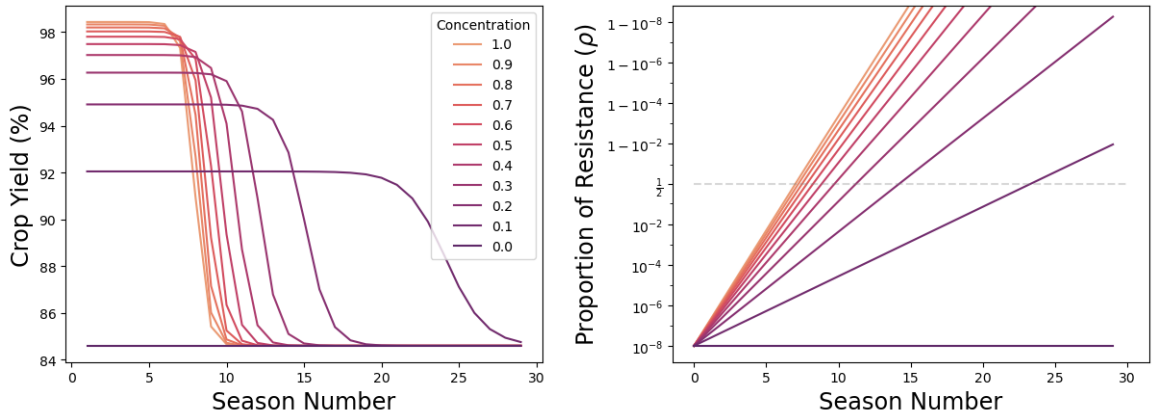
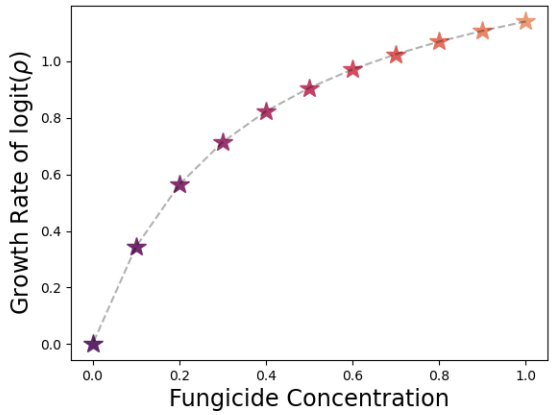


Figure 6: The relationship between fungicide concentration applied and the growth rate of logit(ρ)=$\log_{10}(\frac{\rho}{1-\rho})$.



4.2 Spatially Implicit Multiple Field Model

Between-season ascospore dispersal may cause a significant proportion fungicide resistance genes in the population of Zt to be spread spatially between growing seasons. This results in the rate of resistance selection over multiple seasons to differ from what is expected for fields being considered in isolation due to the possibility of heterogeneity in fungicide usage between fields. To begin to understand this effect quantitatively, growers are divided into two groups based on the dose of fungicide they apply - either using a high fungicide concentration programme with concentration 1 or a low fungicide concentration programme with concentration 0.1. Equations (3a) and (3b) where derived to model the between season ascospore dispersal between the two grower types (see Methods). The proportion of spores incoming from outside a given field is κ and the proportion of high concentration fungicide programmes is q.

$$\rho^{+}_{\text{high}} = (1-\kappa)\rho^{-}_{\text{high}} + \kappa\left(q\rho^{-}_{\text{high}} + (1-q)\rho^{-}_{low}\right) \tag{3a}$$

$$\rho^{+}_{\text{low}} = (1-\kappa)\rho^{-}_{\text{low}} + \kappa\left(q\rho^{-}_{\text{high}} + (1-q)\rho^{-}_{\text{low}}\right) \tag{3b}$$

The spatially implicit multiple field model combines the within-season model (see Figure 3) to independently model the STB epidemic for each grower type (high and low), with between-season ascospore dispersal modelled as $(3a)$ and $(3b)$. The model is simulated with initial proportion of resistance 10^{-8}, for varying parameters κ and q over many years (see Figure 7). The effective lifetime of fungicide (the time, in years, at which the proportion of resistance (ρ) surpasses $1/2$), is recorded for both the low fungicide programme, the high fungicide programme and the difference between these (see Figure 7, top half). Four parameter cases (A, B, C, D) are shown to illustrate the resistance selection dynamics over time for specific κ and q (see Figure 4, bottom half).

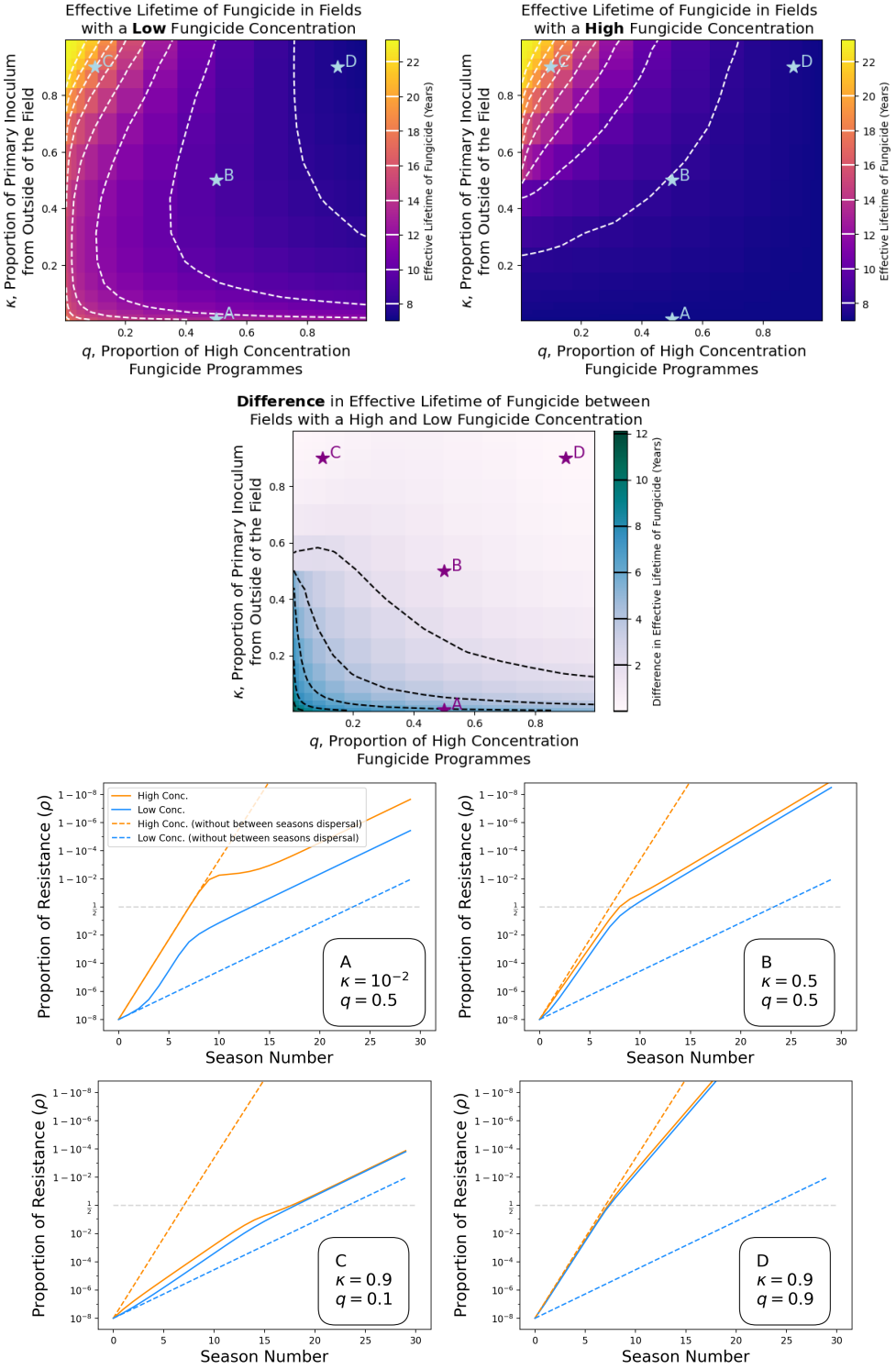
Majority of High Concentration Fungicide Programmes
For $q > 0.5$, the majority of growers are using the high concentration fungicide programme. In this case the effective lifetime of fungicide in fields with the high fungicide concentration remains relatively constant ~ 8 years (the effective lifetime of fungicide considering these fields in isolation) as κ varies (the amount of between season dispersal varies). Hence the rate of resistance selection of a high concentration fungicide programme is relatively unaffected by the presence of a minority of low concentration fungicide programmes. For the low concentration fungicide programmes, the effective lifetime of fungicide monotonically decreases, rapidly at first to ~ 8 as κ increases (See Figure 7, D) (the amount of between season dispersal increases).

Majority of Low Concentration Fungicide Programmes
For $q < 0.5$, the majority of growers are using the low concentration fungicide programme. In this case, the dependence of the effective lifetime of fungicide for the low concentration fungcide programmes on κ is non-linear and exhibits a minimum. As κ increases from 0 the effective lifetime of fungicide decreases to a minimum for low concentration fungicide programmes. This is due to the dispersal of a large amount resistant primary inoculum coming from high concentration fungicide programmes with a large amount also remaining in the fields with high concentration fungicide programmes. As κ increases further, the effective lifetime of fungicide for the low concentration fungicide programmes and high concentration fungicide programmes increases up to ~ 22 years (see Figure 7, C) (similar to the effective lifetime of fungicide of a low concentration fungicide programme as considered in isolation). The high growth rate of the proportion of resistance in high concentration fungicide programmes is being offset through the dilution of the resistant primary inoculum across a majority of low concentration fungicide programmes. Between seasons the proportion of resistant inoculum falls dramatically for the high concentration fungicide programmes, compared to in isolation, and increases marginally for the majority of low concentration fungicide programmes.

For Small κ (low proportion of primary inoculum from originating outside of the field)
From the definition of κ, a proportion equal to κ of primary inoculum originates from outside of the field. Hence the proportion of resistance for the low concentration programme can only grow at a rate different from what it would be considered in isolation up to a proportion of resistance approximately equal to κ. This is seen in case A, with $\kappa = 10^{-2}$. In case A, the proportion of resistance for the low concentration programme grows logistically, over multiple seasons, at the rate of the high concentration programme with 10^{-2} lower magnitude up to $\kappa = 10^{-2}$ in magnitude. Once surpassing $10^{-2}(= \kappa)$, the proportion of resistance of the low concentration programme grows logistically at the rate expected in isolation. One consequence of this is that for $\kappa << 1$, the difference between the effective lifetime of fungicide of the low and high concentration

programmes can only be reduced to the effective lifetime of the low concentration programme with initial proportion of resistance κ, as once surpassing κ the proportion of resistance will grow logistically at the rate expected in isolation.

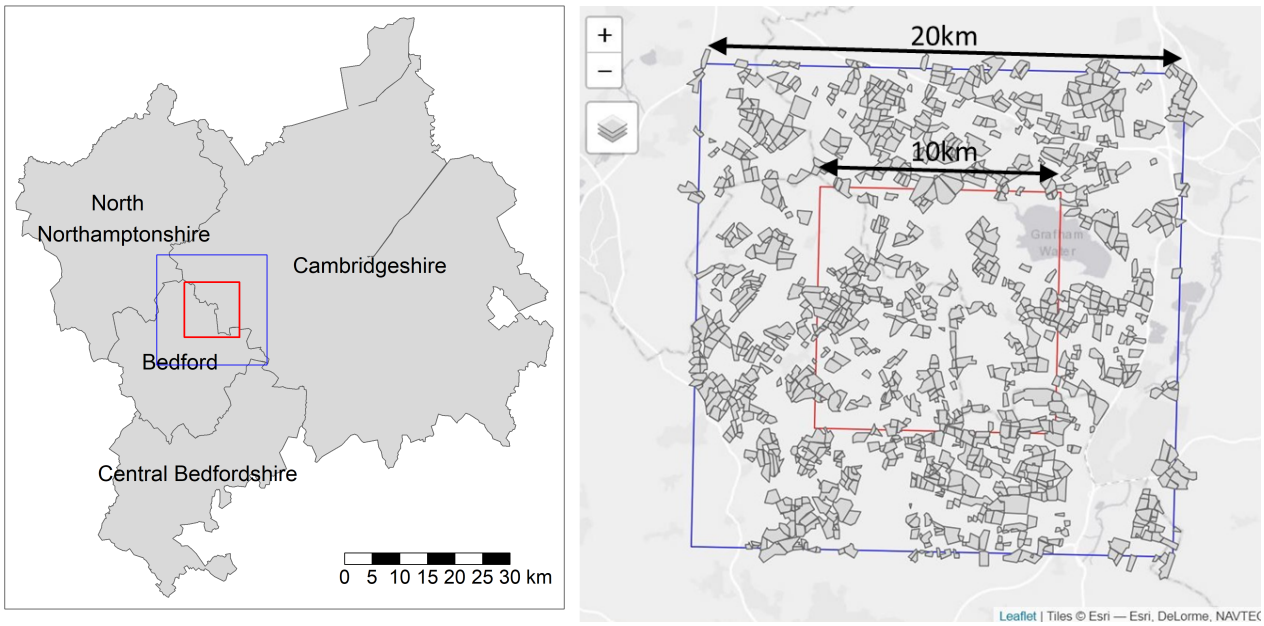
Figure 7: Simulation results of the spatially implicit multiple field model. The dependence of effective lifetime of fungicide on κ and q (top half). How the proportion of resistance grows over time for specific parameter cases of the spatially implicit multiple field model (bottom half)



4.3 Spatially Explicit Multiple Field Model

The spatially implicit multiple field model does not take into account the spatial geometry of fields and between-season dispersal of ascospores. Hence I have constructed a spatially explicit model to take this geometry into account using a real host landscape (see Figure 8) chosen in a region, the East of England, with a large area of winter wheat grown (see Figure 1) and parameterised dispersal kernel respectively (see Methods). In this model, it is assumed the host landscape is static: the location of fields used to grow winter wheat are the same each year (see Discussion).

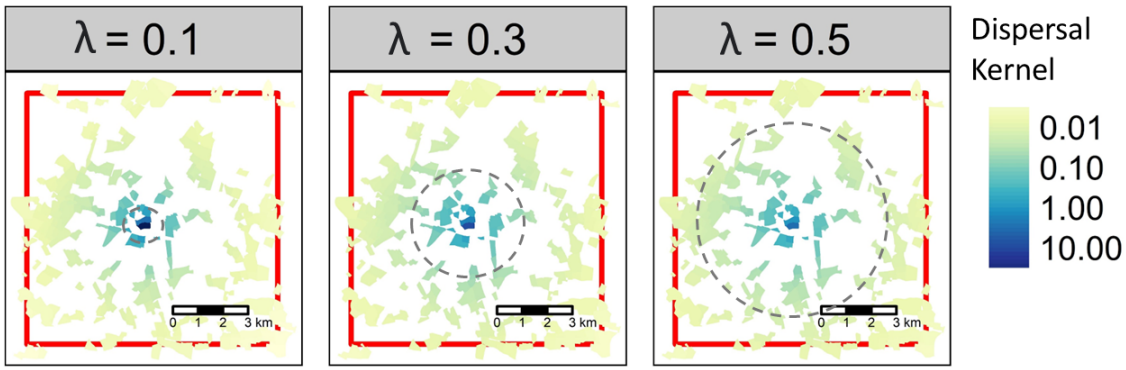
Figure 8: Map of the host landscape, showing its position relative to UK geography and the field polygons with the 10×10km area used in simulations (red) and the 20×20km sourounding area (blue).



To model the between-season dispersal of ascospores, an inverse power law dispersal kernel is used of the form (4) (Karisto, Suffert, & Mikaberidze, 2019). Under certain assumptions, the range of α can be narrowed to $2 < \alpha < 3$ (see Methods). The effect of increasing α or decreasing λ reduces the range of dispersal as defined by the dispersal kernel (4). As α is bounded, I decided to fix $\alpha = 2.5$ and vary λ to increase or decrease the range of dispersal. Throughout, the range of λ considered is $\lambda \in (0.1, 0.5)$ (with d in km) based upon feasible median dispersal, the radius of disc for which 50% of spores are dispersed within (see Figure 9 and Methods). Throughout the units of distance are in km and so the units of λ are km although they may not always be given. In this case λ is thought of as a dimensionless parameter to vary the ascospore dispersal range.

$$p(d) \propto (d + \lambda)^{-\alpha} \tag{4}$$

Figure 9: The value of the dispersal kernel is shown for $\lambda \in \{0.1, 0.3, 0.5\}$, centered on a central field, within the simulation area (red). The disc with radius the median dispersal, is shown with a dashed grey circle, for which 50% of spores from the central field are dispersed.



Throughout i and j are referring to field labels, quantities with i and j subscripts are indicating correspondence to fields i and j. The region of 2D space corresponding to field i is A_i with area $|A_i|$ and d_{ij} is the distance between the centroids of regions A_i and A_j, $d_{ij} = | < A_i > - < A_j > |$. Between-season ascospore dispersal is modelled as in (5) where ρ_i^- and ρ_i^+ are the proportions of the resistance at the end of the previous growing season and the proportion of resistant primary inoculum at the start of the next growing season respectively, for field i (see Methods). Equation (5) is effectively taking ρ_i^+ be be the weighted average of the ρ_j^- with weights $(d_{ij} + \lambda)^{-\alpha}|A_j|$.

$$\rho_i^+ = \frac{\sum_j (d_{ij} + \lambda)^{-\alpha} |A_j| \rho_j^-}{\sum_j (d_{ij} + \lambda)^{-\alpha} |A_j|} \tag{5}$$

Based upon (5), κ_i is defined for field i by (6). This an analogue to κ, the proportion of primary inoculum coming from outside of the field, as used in the spatially implicit model.

$$\kappa_i = 1 - \frac{\lambda^{-\alpha} |A_i|}{\sum_j (d_{ij} + \lambda)^{-\alpha} |A_j|} \tag{6}$$

Unlike the spatially implicit multiple field model, for the spatially explicit multiple field model the value of κ_i varies between fields. The distribution of the κ_i's is shown for varying $\lambda \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$ (with d in km) (see Figure 10, top half). As λ ranges from 0.1 to 0.5 the distribution of κ_i's shifts from being skewed towards 0 to being skewed towards 1. This is reflecting that increasing λ increases the range of between-seasons ascospore dispersal and hence a greater proportion of primary inoculum is coming from outside of the field, increasing κ_i. The order of magnitude of κ_i is on average 10^{-1} to 1 over the range of λ considered. In addition, most of the variation in the κ_i is explained by field area (see Figure 10, bottom half). As the field area of field i increases, the proportion of primary inoculum originating from outside the field decreases, hence decreases κ_i.

The spatially explicit multiple field model combines the within season model (see Figure 3) to independently model the epidemic within each field with between-seasons ascospore dispersal modelled as (5). The model was run over multiple seasons for the 10×10 km host landscape (see Figure 8). Growers are again split into two groups based on the amount of fungicide they apply (as in the spatially implicit multiple field model). Based on the size of the κ_i's (see Figure 10), the

spatially implicit multiple field model suggests that for the spatially explicit multiple field model, the effective lifetime of fungicide for a low concentration fungicide programme will be significantly reduced unless the proportion of high concentration fungicide programmes is low. Hence only one high concentration fungicide programme is modelled at a time with all other fields using a low concentration fungicide programme. By modelling just one, the effect of a single high concentration programme on the effective lifetime of fungicide in terms of the distance from the high concentration programme can be quantified (see Figure 12).

Simulation cases are shown with high concentration fungicide programmes in two different fields (see Figure 11). When field X has the high concentration fungicide programme, increasing the range of between-seasons ascospore dispersal by increasing λ from 0.1 to 0.5 increased the effective lifetime of the fungicide for both grower types to approximately the same value for all fields. However, when field Y has the high concentration fungicide programme, increasing λ from 0.1 to 0.5 decreased the effective lifetime of the fungicide for most fields.

These two opposite effects of increasing the range of between-seasons dispersal on effective lifetime of fungicide can be understood in terms of the field area of the high concentration fungicide programme. A sample of 8 fields was chosen at random, stratified by field area, and for each field in the sample it was chosen to be the high concentration fungicide programme in a simulation of the spatially explicit multiple field model with all other fields low concentration fungicide programmes. The results are shown in Figure 12, showing that as the range of between-seasons ascospore dispersal increases (λ increases from 0.1 to 1) the effective lifetime of fungicide for the low concentration fungicide programmes separate based on the field area of the high concentration fungicide programme, decreasing for area $>\sim$ 40ha and increasing for area $<\sim$ 40ha (see Discussion). Figure 12 also shows how when high concentration fungicide programmes have large field areas, the between-seasons dispersion is unable to offset the high rate of fungicide resistance selection through the dilution of resistant genotype primary inoculum as k_i is low for fields of large areas even when the range of between-seasons ascospore dispersal is increased. This results in the high concentration programme of large field area having low effective lifetime of fungicide for each range of ascospore dispersal considered.

Figure 10: Histograms of the κ_i (the proportion of primary inoculum originating from outside of the field) for each field in the simulation area shown in Figure 8 for $\lambda \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$ (dispersal kernel parameter). (top half) Scatter plots of k_i against the field area showing negative correlation. (bottom half)

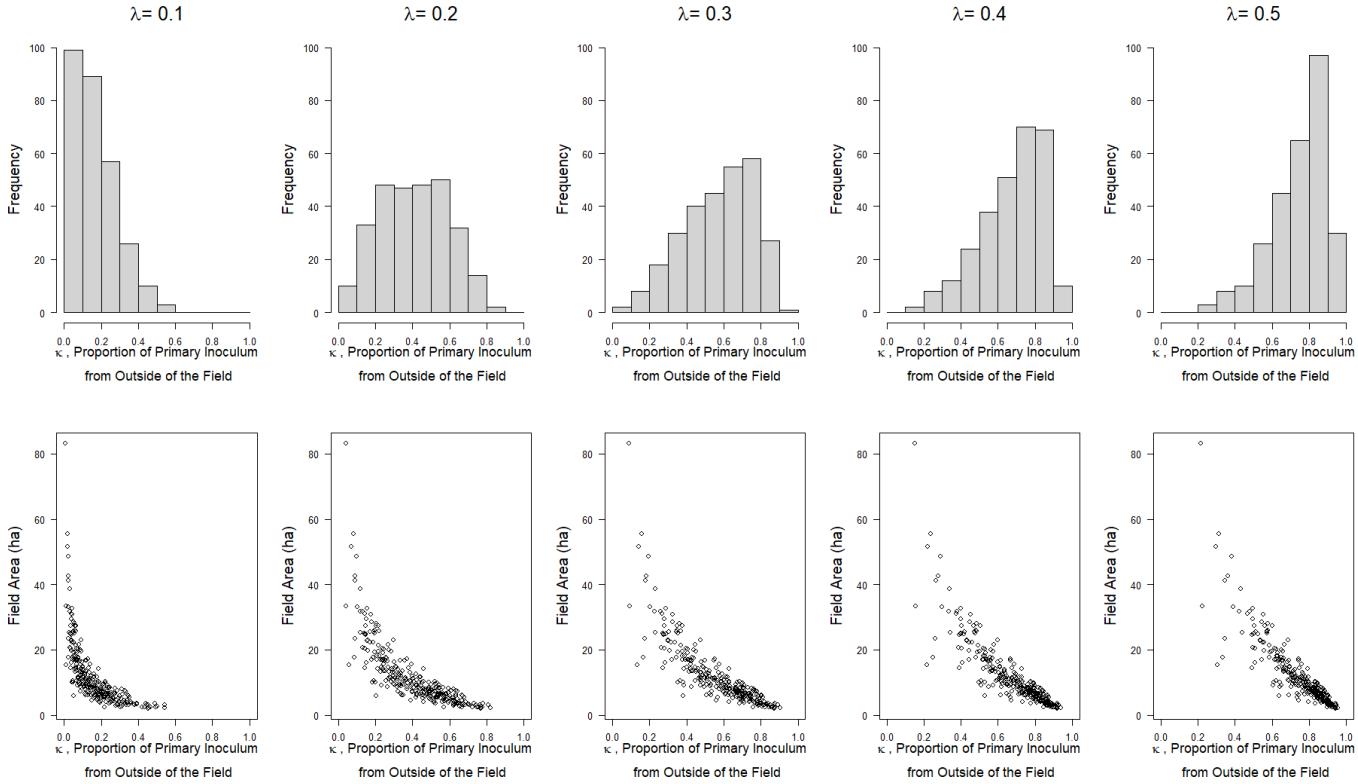


Figure 11: Simulations of the spatially explicit model shown for two different high concentration fungicide programme locations, field X and Y, (orange) with all other growers using a low concentration fungicide programme (blue) for high and low between-seasons dispersal kernel parameter values, $\lambda \in \{0.1, 0.5\}$.

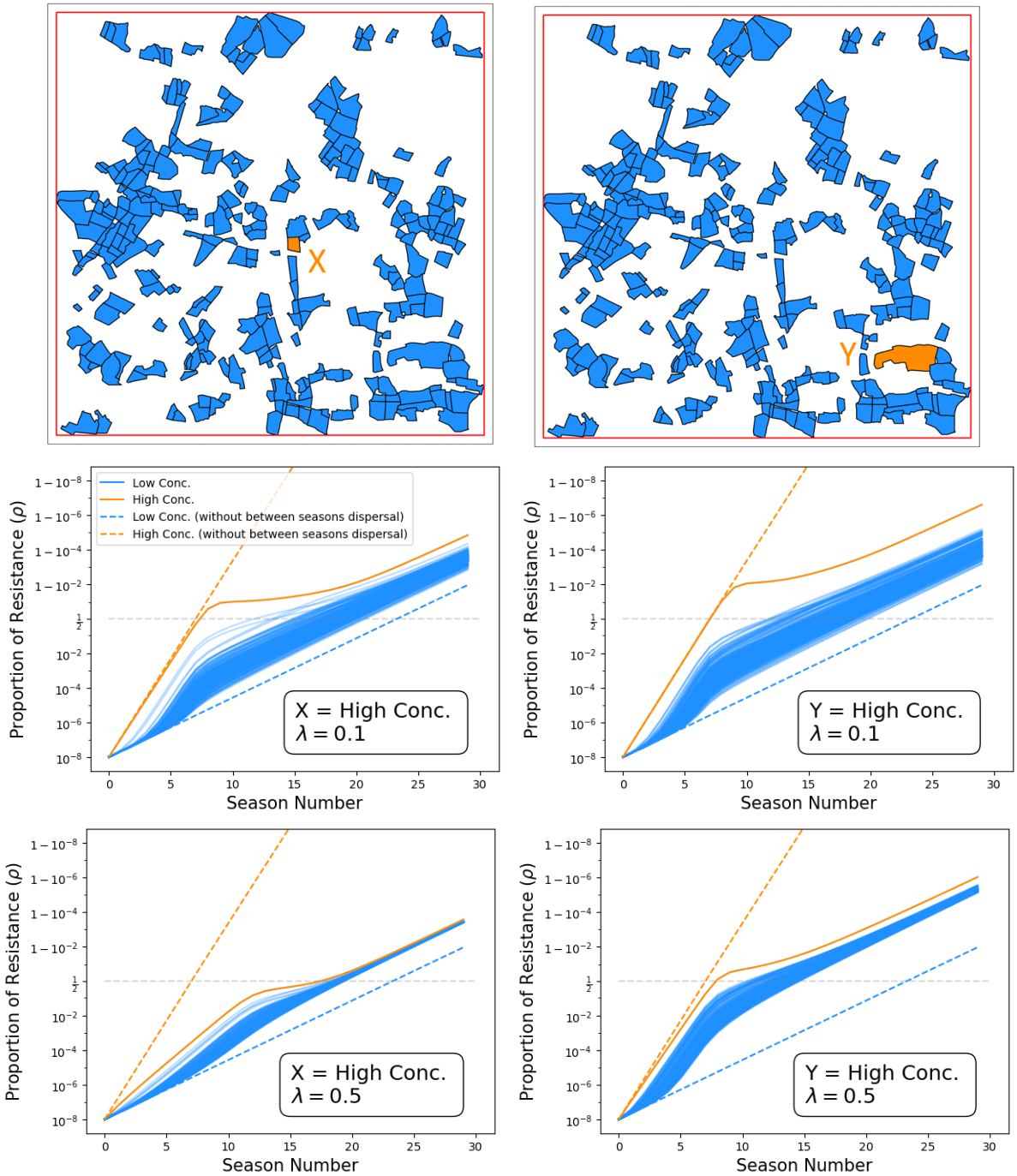
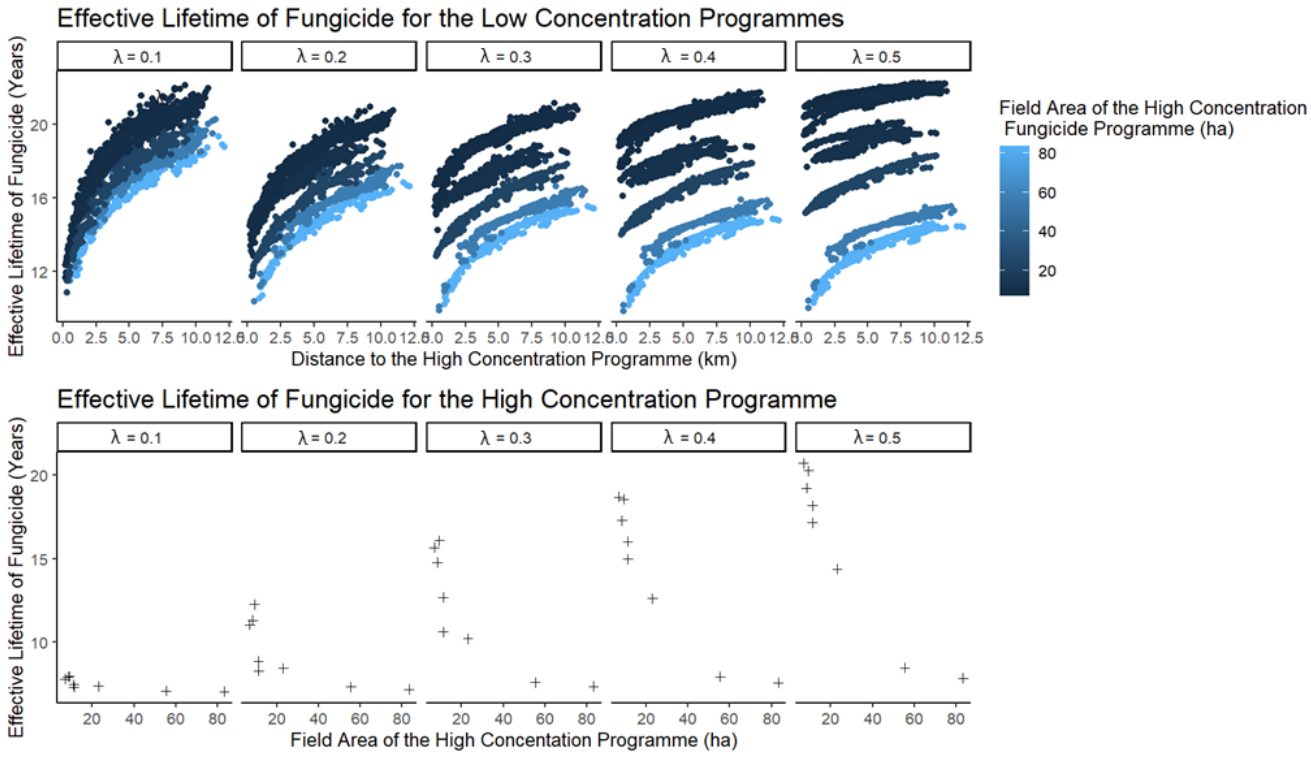


Figure 12: (Top) Effective lifetime of fungicide for the low concentration programmes against the distance to the high concentration fungicide programme coloured by the field area of the high concentration fungicide programme. (Bottom) The corresponding effective lifetime of fungicide for the high concentration programme against the field area for the high concentration programme. (Top and Bottom) For a sample of high concentration fungicide programmes of size 8, stratified by field area for $\lambda \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$ (the dispersal kernel parameter).



5 Discussion

This dissertation has aimed to address the question of how the spatially heterogeneous usage of fungicide between fields and between-seasons ascospore dispersal affects the rate of fungicide resistance selection of *Zt*, over multiple seasons. Overall, the results make clear how fields with high fungicide usage can significantly increase the proportion of fungicide resistance in the population of *Zt* in nearby fields with low fungicide usage, due to between-seasons ascospore dispersal. This means that growers implementing low concentration fungicide programmes nearby to a high concentration fungicide programme will not necessarily benefit from increased effective lifetime of fungicide as predicted by modelling with the assumption of isolation. This necessitates that optimal fungicide resistance management strategies must be rolled out homogeneously across an area to ensure that it is effective over many years.

The spatially implicit multiple field model shows, for a given level of between-seasons ascospore dispersal and a given minimum acceptable effective lifetime of fungicide, there is a maximum threshold for the proportion of high concentration programmes, which is generally very low (see Figure 7). When below this threshold, it is possible that the high proportion of fungicide resistance selected for in fields with high fungicide usage can be offset through the dilution of fungicide resistant genotype primary inoculum across the many fields with low fungicide usage. In addition, when the proportion of high concentration fungicide programmes $< 1/2$, the dependence of the effective lifetime of low concentration fungicide programs on κ is non-linear and exhibits a minimum (See Figure 7).

The proportion of primary inoculum originating from outside of the field (κ) has been shown, in this dissertation, to be an important parameter governing the effect of heterogeneous fungicide usage on the rate of resistance selection. This parameter has been estimated over a real host landscape (see Figure 10), for different feasible ranges of between-seasons ascospore dispersal (see Figure 9). The value of κ varies between fields and I have shown that in dense host landscapes, such as the one considered in this dissertation, the field area has a strong negative correlation with κ (see Figure 10). The spatially explicit multiple field model shows that for fungicide resistant genotype primary inoculum to be diluted away from high fungicide usage areas to low fungicide usage areas, the area of fields using high fungicide concentration programmes must be sufficiently low, else κ is low and dilution is not possible. Hence large field areas (> 40ha) with high fungicide usage cause the effective lifetime of fungicide to be significantly reduced in nearby fields with low fungicide usage. This reduction increases as the range of between field dispersion increases, meaning the dilution of resistant genotype primary inoculum is not feasible for high concentration fungicide programmes in fields with large areas.

The spatially explicit multiple field model shows, for a low concentration fungicide programme to benefit from increased effective fungicide lifetime, the field must be sufficiently far, at least several kilometres, from a high concentration fungicide programme, or alternatively, if the distance is less the field with a high fungicide usage must have a sufficiently low area and the range of between-seasons ascospore dispersal must be sufficiently large, surrounded by a high density of winter wheat fields with low fungicide usage.

The implications of this to fungicide resistance management is that to successfully decrease the rate of fungicide resistance selection through reduced dependence on and use of fungicides, fungicide usage must be locally coordinated to achieve homogenous low levels of fungicide usage. Moreover, to achieve a larger effective lifetime for fungicides through reducing fungicide usage it is important that the fields of the largest areas are applying reduced fungicide concentrations. Area wide management is often used to help control plant diseases (Garcia Figuera, Babcock, Lubell, & McRoberts, 2022). Further work could involve finding ways to optimally foster collective action in the management of fungicide resistance and reduce the dependence on fungicides in agriculture.

The main uncertainty in these conclusions arises from the uncertainty in the form and parameterisation of the dispersal kernel for between-seasons ascospore dispersal as well as the validity of the assumption of uniform disease risk across the host landscape and between years. The parameter λ in the inverse power law dispersal kernel used has been varied while fixing α to consider different ranges of dispersal. The range of λ has been chosen such that the median dispersal of the dispersal kernel is feasible. There are also uncertainties in the within-season model structure and parameterisation as in (Elderfield et al., 2018; Hobbelen et al., 2011).

Limitations of the methods used in the work include using a static host landscape over multiple years for the spatially explicit model as opposed to a dynamic host landscape. Between years the fields used to grow winter wheat change due to crop rotations (see Figure 13). However, the density of fields remains similar over time so the conclusions of this work may still be valid in this case. Another limitation is that fungicide concentration is assumed to be the same each year over multiple seasons. Further work could incorporate a dynamic host landscape and variable fungicide usage over time to understand how a dynamic host landscape and variable fungicide usage can be used in a fungicide resistance management strategy to reduce the rate of fungicide resistance development over an area. Further work could also be done to not use the approximation used to compute the probability of spores travelling from one field to another in the spatially explicit model and instead computing the exact integral over both fields (see Methods)

Further work can be done to incorporate the between seasons sexual reproduction of Zt as in (Taylor & Cunniffe, 2022) and in addition spatial heterogeneity in the usage of multiple fungicides with different modes of action as well as fungicide mixtures (Elderfield et al., 2018). In the case of resistance to multiple fungicides, sexual reproduction will increase rate of resistance selection to multiple fungicides as sexual reproduction may occur between Zt exposed to fungicides of different modes of action. This further suggests the need for local coordination in fungicide resistance management.

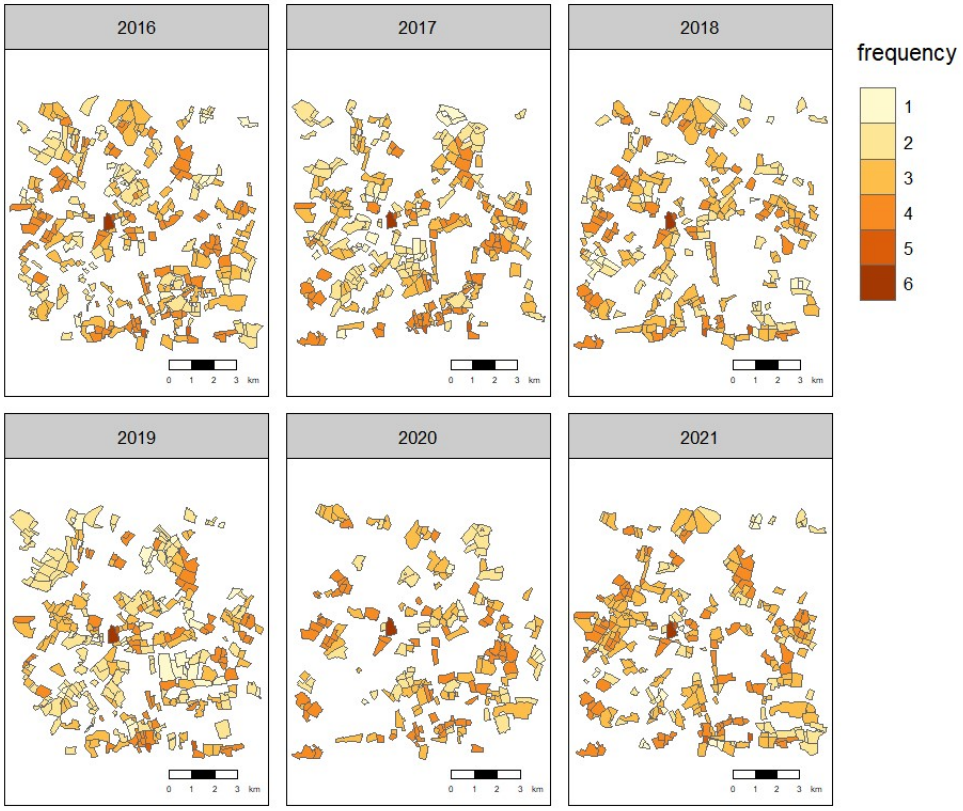
In conclusion, to address the question of how the success of a fungicide resistance management strategy responds to a proportion of growers who do not follow recommendations. The following has been observed:

- For a given range of between-seasons ascospore dispersal and a given minimum acceptable effective lifetime of fungicide, there is a maximum threshold for the proportion of high concentration fungicide programmes.
- The spatially implicit multiple field model shows, when the proportion of high concentration fungicide programmes $< 1/2$, the dependence of the effective lifetime of fungicide for low

concentration fungicide programmes on κ is non-linear and exhibits a minimum (see Figure 7).

- There is a threshold for the field area of a high concentration fungicide programme, such that beyond which the proportion of fungicide resistant genotype primary inoculum originating from the field cannot be diluted by a large number of surrounding fields with low concentration fungicide programmes to an extent to substantially slow down the rate of resistance selection.

Figure 13: How the host landscape varies over the years 2016-2021 coloured by the frequency of how many times fields are used to grow winter wheat over the 6 years.



6 Methods

6.1 Single Field Model

To model the epidemic within a season a compartmental model is used based on (Elderfield et al., 2018; Hobbelen et al., 2011) (see Figure 3). The compartments correspond to the Leaf Area Index (LAI), the area of leaf per area of ground, of different types of leaf tissue, Susceptible (S), Exposed (E), Infected (I) or Removed (R). Both compartments E and I are split into disease that is due fungicide resistant or fungicide susceptible Zt respectively, E_r and E_s, I_r and I_s. Primary inoculum corresponds to P_r and P_s, fungicide susceptible and resistant respectively, which initiate the epidemic in the early stage of the growing season. The concentration of fungicide is C, with fungicide sprayed at growth stages 32 and 39 (see Table 2). Unless specified otherwise, when fungicide concentration is referred to throughout this text it is referring to the concentration sprayed at these times, which are assumed to be equal. This concentration ranges from 0 to a full dose 1.

The within-season model starts at the time of emergence of leaf 5 (T_{EMERGE}) and ends at growth stage 87 (GS87) (see Table 2). The within-season model equations (11-19) are solved using the ode solver implemented in the Python package SciPy (Virtanen et al., 2020).

To simulate the selection of resistance over multiple growing seasons, with within-season model is run recursively. The end condition for the proportion of resistance as defined by (1), $\rho^- = \rho(T_{\text{GS87}})$ of the previous season becomes the initial resistance proportion for the next season $\rho^+ = \rho(T_{\text{EMERGE}})$.

The total leaf area index (LAI) given by,

$$A = S + E_r + E_s + I_r + I_s + R \tag{7}$$

Host growth is given by,

$$g(A) = \begin{cases} 0, & t < T_{\text{EMERGE}} \\ r(k - A), & t \geq T_{\text{EMERGE}} \end{cases} \tag{8}$$

Host senescence is given by,

$$\Gamma(A) = \begin{cases} 0, & t < T_{\text{GS61}} \\ 0.005 \left(\frac{t - T_{\text{GS61}}}{(T_{\text{GS87}} - T_{\text{GS61}})} \right) + 0.1 e^{-0.02(T_{\text{GS87}} - t)}, & t \geq T_{\text{GS61}} \end{cases} \tag{9}$$

Effect of fungicide is given by,

$$\epsilon(C) = \omega(1 - e^{\theta C}) \tag{10}$$

Within-season model equations:

$$\frac{dS}{dt} = g(A) - \Gamma(t)S - \frac{\beta S}{A}((1-\epsilon(C))(I_S + P_s) + I_r + P_r) \tag{11}$$

$$\frac{dE_r}{dt} = \frac{\beta S}{A}(1-\epsilon(C))(I_r + P_r) - \Gamma(t)E_r - \gamma E_r \tag{12}$$

$$\frac{dE_s}{dt} = \frac{\beta S}{A}(1-\epsilon(C))(I_s + P_s) - \Gamma(t)E_s - \gamma E_s \tag{13}$$

$$\frac{dI_r}{dt} = \gamma E_r - \mu I_r \tag{14}$$

$$\frac{dI_s}{dt} = \gamma E_s - \mu I_s \tag{15}$$

$$\frac{dR}{dt} = \mu(I_s + I_r) + \Gamma(t)(S + E_r + E_s) \tag{16}$$

$$\frac{dP_r}{dt} = -\upsilon P_r \tag{17}$$

$$\frac{dP_s}{dt} = -\upsilon P_s \tag{18}$$

$$\frac{dC}{dt} = -\delta C \tag{19}$$

See Table 2 for all the within-season model parameter values.

Table 2: Table of parameters used in the within-season model, based on (Elderfield et al., 2018; Hobbelen et al., 2011)

Parameter	Symbol	Value
Infection Rate	β	1.56×10^{-2} degree-days^{-1}
Latent period	$1/\gamma$	266 degree-days
Infectious period	$1/\mu$	456 degree-days
Fungicide maximum effect	ω	1
Fungicide dose response curvature	θ	9.6
Fungcide decay parameter	δ	1.11×10^{-2}degree-days^{-1}
Initial resistance proportion	ρ^+	$\in [0, 1]$
Initial inoculum density	ϕ	$1{\times}10^{-10}$
Initial Leaf Area Index (LAI) of susceptible leaf	$S(T_{\text{EMERGE}})$	0.05 m^{2}m^{-2}
Initial LAI of leaf with latent infection by susceptible strain	$E_s(T_{\text{EMERGE}})$	0 m^{2}m^{-2}
Initial LAI of leaf with latent infection by resistant strain	$E_r(T_{\text{EMERGE}})$	0 m^{2}m^{-2}
Initial LAI of leaf infected by susceptible strain	$I_s(T_{\text{EMERGE}})$	0 m^{2}m^{-2}
Initial LAI of leaf infected by resistant strain	$I_s(T_{\text{EMERGE}})$	0 m^{2}m^{-2}
Initial LAI of dead leaf infected	$R(T_{\text{EMERGE}})$	0 m^{2}m^{-2}
Host growth rate parameter	r	8.5×10^{-3}degree-days^{-1}
Host carrying capacity	k	$4.2\text{m}^2 m^{-2}$
Decay rate of primary inoculum	v	8.5×10^{-3}degree days^{-1}
Spray times	-	T_{GS32},T_{GS39}
Time of emergence of leaf 5	T_{EMERGE}	1212 degree days
Time of GS32	T_{GS32}	1212 degree days
Time of GS39	T_{GS39}	1700 degree days
Time of GS61	T_{GS61}	2066 degree days
Time of GS87	T_{GS87}	2900 degree days
Initial LAI of lower leaf infected by resistant strain	$\text{P}_r(T_{\text{EMERGE}})$	$\phi\rho^+$
Initial LAI of lower leaf infected by susceptible strain	$\text{P}_r(T_{\text{EMERGE}})$	$\phi(1-\rho^+)$

6.2 Spatially Implicit Multiple Field Model

The spatially implicit multiple field model growers are divided into two groups, growers who are spraying a high concentration of fungicide and growers who are spraying a lower concentration respectively. I have taken the high concentration to be the full dose $C = 1$ and the low dose to be $C = 0.1$ the reason for this being that the difference is larger enough that the effect has the best chance of being quantified and hence understood.

Taking q =the proportion of high concentration fungicide programmes and κ =the proportion of primary inoculum originating from outside of the field. Under the assumption of homogeneity of ascospores in the environment originating from fields from high and low concentration fungicide programmes, in the respective proportions and the proportion of resistant genotype primary inoculum for growers of the same type is equal. Between-seasons ascospore dispersal can be modelled as:

$$\rho^{+}_{\text{high}} = (1-\kappa)\rho^{-}_{\text{high}} + \kappa\left(q\rho^{-}_{\text{high}} + (1-q)\rho^{-}_{low}\right) \tag{20a}$$

$$\rho^{+}_{\text{low}} = (1-\kappa)\rho^{-}_{\text{low}} + \kappa\left(q\rho^{-}_{\text{high}} + (1-q)\rho^{-}_{\text{low}}\right) \tag{20b}$$

The spatially implicit multiple field model consists of modelling the fields of each grower type independently with the within-season model as in the single field model. Then computing the proportion of resistance at the end of the season for the fields of each grower type. The initial proportion of resistant genotype primary inoculum for each grower type in the next year is given by equations (20a) and (20b).

6.3 Spatially Explicit Multiple Field Model

The host landscape used for the spatially explicit multiple field model and Figure 1 are based upon UKCEH Land Cover® Plus: Crops © [2021] UKCEH. © RSAC. © Crown Copyright 2007, Licence number 100017572. This data set gives the locations and geometry of over 200,000 fields used to grow winter wheat across the UK for each of the years 2016 to 2021, detected through satellite imagery and agricultural census. This corresponds to virtually all fields of winter wheat across the UK. A subset of this data, for 2021, consisting of 284 fields in a 10×10 km area on the Cambridgeshire-Bedfordshire border is used as a real host landscape for the spatially explicit multiple field model. This scale of host landscape is appropriate for this study as both between-season ascospore dispersal has been observed (Shaw & Royle, 1989) at this scale and heterogeneity in fungicide usage is likely to be significant this scale.

I have used the inverse power law dispersal kernel (Karisto et al., 2019) as in equation (4) to model the between season dispersal of ascospores. To bound the parameter α in the inverse power law dispersal kernel $p(d) = (d+\lambda)^{-\alpha}$, let μ_0 and μ_1 be the zeroth and first moments of p. As $p(d)$ is a probability density over 2 dimensional space μ_0 converges ($\mu_0 < \infty$), this gives the constraint $\alpha > 2$. Since the range of the airborne ascospore dispersal is known to take place over multiple scales of magnitude from meters to kilometers, $p(d)$ is fat tailed, hence we can speculate the first moment is divergent, $\mu_1 = \infty$ this gives the condition $\alpha < 3$. Hence α is estimated to be within the interval $(2, 3)$.

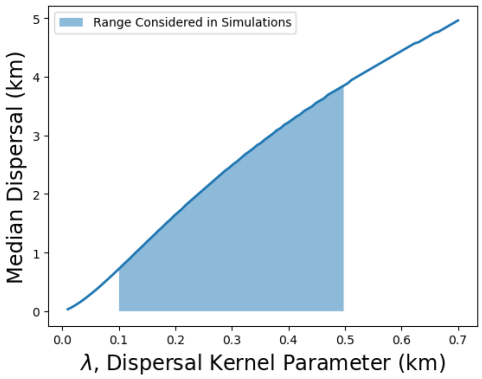
As α is bounded, I decided to fix $\alpha = 2.5$ and vary λ to increase or decrease the range of dispersal. To estimate a feasible range for λ, since the first moment of the dispersal kernel diverges, the

median dispersal is defined as the radius the of disc for which 50% of ascospores are dispersed within. The median dispersal d_{med} can be found by solving the equation,

$$\frac{\int_0^{2\pi}\int_0^{d_{med}} \frac{r}{(d+\lambda)^{\alpha}} dr d\theta}{\int_0^{2\pi}\int_0^{\infty} \frac{r}{(d+\lambda)^{\alpha}} dr d\theta} = 1/2 \tag{21}$$

Equation (21) is solved numerically for d_{med} using the bisection method for given λ (see Figure 14). As observed in (Shaw & Royle, 1989) STB infections in winter wheat where regularly found a site 0.4 km distant from wheat due to airborne ascospores and ascospores are known to regularly travel much further based on the genetic diversity of populations of *Zt* (McDonald, Suffert, Bernasconi, & Mikaberidze, 2022). Hence it can be speculated that the median dispersal of ascospores is > 0.4km. With this and from the relationship of d_{med} and λ (see Figure 14), a feasible range of $\lambda \in [0.1, 0.5]$km is used in the spatially explicit multiple field model.

Figure 14: Median Dispersal as a function of the Dispersal Kernel Parameter λ. The range of $\lambda \in [0.1, 0.5]$ considered in the simulations is indicated by solid blue region under the graph.



Let A_i denote the region corresponding to the ith field in the area of interest and $(A_j \to A_i)$ the event of a spore originating from field Aj reaching A_i as primary inoculum. Additionally let $< a_i >$ be the centroid of field A_i with $d_{ij} = | < a_i > - < a_j > |$. Then the probability of $(A_j \to A_i)$ can be estimated by (26). Equation (22) assumes that disease risk is spatially uniform and hence ascospore production is uniform across all fields. The approximation (23) is not always valid but has been made for computation simplicity, further work could do similar modelling without using this approximation.

$$P(A_j \to A_i) = \int_{a_i \in A_i}\int_{a_j \in A_j} p(|a_i - a_j|) dA_i dA_j \tag{22}$$

$$\propto \int_{a_i \in A_i}\int_{a_j \in A_j} (|a_i - a_j| + \lambda)^{-\alpha} \, dA_i dA_j \tag{23}$$

$$\approx (|\langle a_i \rangle - \langle a_j \rangle| + \lambda)^{-\alpha} \, |A_i||A_j| \tag{24}$$

$$= (d_{ij} + \lambda)^{-\alpha} |A_i||A_j| \tag{25}$$

$$\therefore P(A_j \to A_i) \propto\!\!\!\!\sim (d_{ij} + \lambda)^{-\alpha} |A_i||A_j| \tag{26}$$

The proportion of resistant genotype primary inoculum in field i (ρ_i^+) is a weighted sum of the proportion of resistance at the end of the previous growing season in each field in the surrounding

area (ρ_j^-). The maximum likehood estimate of ρ_i^+ corresponds to these weights being proportional to the probabilities $P(A_j \to A_i)$. Hence,

$$\rho_i^+ = \frac{\sum_j P(A_j \to A_i)\rho_j^-}{\sum_j P(A_j \to A_i)} = \frac{\sum_j (d_{ij} + \lambda)^{-\alpha}|A_j|\rho_j^-}{\sum_j (d_{ij} + \lambda)^{-\alpha}|A_j|} \tag{27}$$

Based upon (27), κ_i is defined for field i by (28). This an analogue to κ, the proportion of primary inoculum coming from outside of the field, as used in the spatially implicit model.

$$\kappa_i = 1 - \frac{\lambda^{-\alpha}|A_i|}{\sum_j (d_{ij} + \lambda)^{-\alpha}|A_j|} \tag{28}$$

The spatially explicit multiple field model consists of modelling each field independently with the within-seasons single field model, with the respective concentration of fungicide. Then computing the proportion of resistance at the end of the season for each field. The initial proportion of resistant genotype primary inoculum in the next year is given by (27). The spatially explicit model assumes uniform disease risk with uniform production of ascospores across fields and that primary inoculum is entirely composed of ascospores.

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